

Pair of Straight line - Chap-5

General eqn of 2nd degree

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$$

General eqn of pair of straight line

$$(x-y)(x-y) \quad ax^2 + 2hxy + by^2 = 0$$

$$\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$$

if, $\Delta = 0$, or, $abc + 2fgh - af^2 - bg^2 - ch^2 = 0$
then it is a pair of straight line eqn.

Intersection point

$$\alpha = \frac{hf - bg}{ab - h^2}, \quad \beta = \frac{gh - af}{ab - h^2}$$

Angle $\tan \theta = \frac{2\sqrt{h^2 - ab}}{a+b}$

↳ condition of perpendicular $a+b = 0$

↳ condition of parallel $h^2 - ab = 0$
 $\Rightarrow h^2 = ab$

The bisector of the angles betⁿ the lines
of $ax^2 + 2hxy + by^2 = 0$

is, $\frac{x^2 - y^2}{a - b} = \frac{xy}{h}$

for, General eqn,

$$\frac{(x - \alpha)^2 - (y - \beta)^2}{a - b} = \frac{(x - \alpha)(y - \beta)}{h}$$

$0 = (x - \alpha)^2 - (y - \beta)^2 - \frac{(x - \alpha)(y - \beta)(a - b)}{h}$

$$\frac{bx - cy}{a - b} = h \quad \frac{cy - bx}{a - b} = h$$

$$\frac{bx - cy}{a - b} = h \quad \frac{cy - bx}{a - b} = h$$

$0 = (x - \alpha)^2 - (y - \beta)^2 - \frac{(x - \alpha)(y - \beta)(a - b)}{h}$

$0 < (x - \alpha)^2 - (y - \beta)^2 - \frac{(x - \alpha)(y - \beta)(a - b)}{h}$

$0 > (x - \alpha)^2 - (y - \beta)^2 - \frac{(x - \alpha)(y - \beta)(a - b)}{h}$

$$x+y = n$$

$$y = mx + c$$

Exercise Chap-5

Find the lines represented by

i. $x^2 \cos 2\theta - 4xy \cos \theta + 2y^2 + x^2 = 0$

ii. $3x^2 - 16xy + 5y^2 = 0$

iii. $y^3 - xy^2 - 14x^2y + 24x^3 = 0$

Soln:

i. $x^2 \cos 2\theta - 4xy \cos \theta + 2y^2 + x^2 = 0$

$$\Rightarrow x^2 (\cos 2\theta + 1) - 4xy \cos \theta + 2y^2 = 0$$

$$\Rightarrow x^2 (2\cos^2 \theta) - 4xy \cos \theta + 2y^2 = 0$$

$$\Rightarrow 2(x^2 \cos^2 \theta - 2xy \cos \theta + y^2) = 0$$

$$\Rightarrow x^2 \cos^2 \theta - 2xy \cos \theta + y^2 = 0$$

$$\Rightarrow (x \cos \theta - y)^2 = 0$$

$$\Rightarrow \underline{y = x \cos \theta} \quad \underline{\text{Ans}} \quad \underline{y = x \cos \theta} \quad \underline{\text{Ans}}$$

$$\textcircled{II} 3x^2 - 16xy + 5y^2 = 0$$

$$\Rightarrow 3x^2 - 15xy - xy + 5y^2 = 0$$

$$\Rightarrow 3x(x - 5y) - y(x - 5y) = 0$$

$$\Rightarrow (x - 5y)(3x - y) = 0$$

$$\therefore x - 5y = 0 \quad \text{or} \quad 3x - y = 0 \quad \underline{\text{Ans}}$$

$$\textcircled{III} y^3 - xy^2 - 14x^2y + 24x^3 = 0$$

$$\Rightarrow y^3 - 2xy^2 + xy^2 - 2x^2y - 12x^2y + 24x^3 = 0$$

$$\Rightarrow y^2(y - 2x) + xy(y - 2x) - 12x^2(y - 2x) = 0$$

$$\Rightarrow (y - 2x)(y^2 - xy - 12x^2) = 0$$

$$\text{Now, } y^2 - xy - 12x^2 = 0$$

$$\Rightarrow y^2 - 3xy + 4xy - 12x^2 = 0$$

$$\Rightarrow y(y - 3x) + 4x(y - 3x) = 0$$

$$\Rightarrow (y - 3x)(y + 4x) = 0$$

$$\therefore (y - 2x)(y - 3x)(y + 4x) = 0$$

~~$y = 2x$~~
 ~~$3x - 4x = 3$~~
 ~~$-12x^2 + 24x^3$~~

~~$y = 3x$~~
 ~~$y = 4x$~~

Ans

prove that the following eqn represent two straight line, find also their point of intersection and the angles between them,

20

$$IV. 3y^2 - 8xy - 3x^2 - 29x + 3y - 18 = 0$$

$$V. 2y^2 - 2y - x^2 + y + 2x - 1 = 0$$

$$VI. x^2 - 5xy - 6y^2 + 14x + 5y + 4 = 0$$

$$VII. 2y^2 + 3xy - 5y - 6x + 2 = 0$$

Soln:

$$IV. 3y^2 - 8xy - 3x^2 - 29x + 3y - 18 = 0$$

$$\Rightarrow -3x^2 - 8xy + 3y^2 - 29x + 3y - 18 = 0 \quad \text{--- (1)}$$

comparing (1) with $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

$$a = -3, b = 3, h = -4, g = -\frac{29}{2}, f = \frac{3}{2}, c = -18$$

intersection point (α, β)

$$\alpha = \frac{hf - bg}{ab - h^2}, \quad \beta = \frac{gh - af}{ab - h^2}$$

$$= \frac{(-4 \times \frac{3}{2}) - (3 \times -\frac{29}{2})}{(-3 \times 3) - (-4)^2}$$

$$= \frac{-6 + \frac{87}{2}}{-9 - 16}$$

$$= \frac{\frac{75}{2}}{-25} = -\frac{3}{2}$$

2

$$= \frac{(-\frac{29}{2} \times -4) - (-3 \times \frac{3}{2})}{(-3 \times 3) - (-4)^2}$$

$$= \frac{58 + \frac{9}{2}}{-25}$$

$$= -\left(\frac{125}{2} \times \frac{1}{25}\right)$$

$$= -\frac{5}{2}$$

intersection point $(x, y) = (-3/2, -5/2)$

$$\tan \theta = \frac{2\sqrt{h^2 - ab}}{a+b}$$

$$= \frac{2\sqrt{(1)^2 - (-3) \cdot 3}}{-3+3}$$

$$= \frac{2 \cdot 5}{0}$$

$$\tan \theta = \infty$$

$$\therefore \theta = \tan^{-1}(\infty)$$

$$= 90^\circ$$

(90°) being perpendicular

$$\frac{bx - dy}{x - dy} = d$$

$$\frac{cd - bx}{x - dy} = c$$

$$\left(\frac{0}{1} \times \frac{20}{1} \right) - \left(\frac{10}{1} \times \frac{0}{1} \right) =$$

$$\frac{0 + 20}{1 - 0} =$$

$$\left(\frac{1}{20} \times \frac{20}{1} \right) -$$

$$\frac{10}{1} =$$

$$\left(\frac{10}{1} \times \frac{0}{1} \right) - \left(\frac{0}{1} \times \frac{10}{1} \right) =$$

$$\frac{0 - 10}{1 - 0} =$$

$$\frac{10}{1} + 2 =$$

$$12 =$$

$$\left(\frac{1}{20} \times \frac{20}{1} \right) -$$

$$\frac{10}{1} =$$

$$\textcircled{v} 2y^2 - xy - x^2 + y + 2x - 1 = 0$$

$$\Rightarrow -x^2 - xy + 2y^2 + 2x + y - 1 = 0 \quad \text{--- (1) comparing (1)}$$

$$\text{with } ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$$

$$a = -1, h = -\frac{1}{2}, b = 2, g = 1, f = \frac{1}{2}, c = -1$$

intersection point,

$$\begin{aligned} \alpha &= \frac{hf - bg}{ab - h^2} \\ &= \frac{-\frac{1}{2} \times \frac{1}{2} - 2 \times 1}{-2 - \left(-\frac{1}{2}\right)^2} \\ &= \frac{-\frac{1}{4} - 2}{-2 - \frac{1}{4}} \\ &= \frac{-\frac{9}{4}}{-\frac{9}{4}} \\ &= 1 \end{aligned}$$

$$\begin{aligned} \beta &= \frac{hg - af}{ab - h^2} \\ &= \frac{\left(-\frac{1}{2} \times 1\right) - \left(-1 \times \frac{1}{2}\right)}{-2 - \left(-\frac{1}{2}\right)^2} \\ &= \frac{-\frac{1}{2} + \frac{1}{2}}{-\frac{9}{4}} \\ &= 0 \end{aligned}$$

$$\text{intersection point } (\alpha, \beta) = (1, 0)$$

Ans

$$\begin{aligned} \tan \theta &= \frac{2\sqrt{h^2 - ab}}{a + b} \\ &= \frac{2\sqrt{\left(-\frac{1}{2}\right)^2 - (-1 \times 2)}}{-1 + 2} \\ &= \frac{2\sqrt{\frac{1}{4} + 2}}{1} \end{aligned}$$

$$\tan \theta = -3$$

$$\Rightarrow \theta = \tan^{-1}(-3)$$

Ans

આથરે હાલ સમજાવે $4 \neq 0$ 50, યોગ્ય પાત્ર
of straight line
સરખા

(VI) $x^2 - 5xy - 6y^2 + 14x + 5y + 4 = 0$ — (1)

comparing with $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

$a = 1, h = -5/2, b = -6, g = 7, f = 5/2, c = 4$

$\Delta = \frac{hf - bg}{ab - h^2}$

$= \frac{(-5/2 \times 5/2) - (-6 \times 7)}{(1 \times -6) - (-5/2)^2}$

$= \frac{-25/4 + 35}{-6 - 25/4}$

$= \frac{-25/4 + 35}{-6 - 25/4}$

$= \frac{115/4}{-37/4}$

$$\textcircled{\text{vii}} \quad 2y^2 + 3xy - 5y - 6x + 2 = 0$$

$$\Rightarrow 3xy + 2y^2 - 6x - 5y + 2 = 0 \quad \text{--- (1)}$$

compare (1) with $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

$$a=0, \quad h=3/2, \quad b=2, \quad g=-3, \quad f=-5/2, \quad c=2$$

$$\alpha = \frac{hf - bg}{ab - h^2} = \frac{(3/2 \times -5/2) - (2 \times -3)}{0 \times 2 - (3/2)^2} = \frac{-\frac{15}{4} + 6}{-\frac{9}{4}} = -\left(\frac{9}{4} \times \frac{4}{9}\right) = -1$$

$$\beta = \frac{hg - af}{ab - h^2} = \frac{(3/2 \times -3) - (0 \times -5/2)}{0 \times 2 - (3/2)^2} = \frac{-\frac{9}{2}}{-\frac{9}{4}} = \frac{2}{1} = 2$$

\(\therefore\) intersection point \((\alpha, \beta) = (-1, 2)\)

$$\tan \theta = \frac{2\sqrt{h^2 - ab}}{a + b}$$

$$= \frac{2\sqrt{(3/2)^2 - 0 \times 2}}{0 + 2}$$

$$= \frac{2 \times \frac{3}{2}}{2}$$

$$\Rightarrow \tan \theta = \frac{3}{2}$$

$$\Rightarrow \theta = \tan^{-1}\left(\frac{3}{2}\right)$$

Find the value of λ on K

$$\text{viii. } \lambda x^2 + 4xy + y^2 - 4x - 2y - 3 = 0$$

$$\text{ix. } 2x^2 - y^2 + xy - 2x - 5y + k = 0$$

$$\text{x. } 6x^2 + xy + ky^2 - 11x + 43y - 35 = 0$$

$$\text{xi. } x^2 - \lambda xy + 2y^2 + 3x - 5y + 2 = 0$$

$$\text{xii. } kxy - 8x + 9y - 12 = 0$$

$$\text{xiii. } 6x^2 + 2kxy + 12y^2 + 22x + 31y + 20 = 0$$

~~$$\text{xiv. } 6x^2 - 7xy - 3y^2 - 2y + k = 0$$~~

$$\text{xv. } 12x^2 - 10xy + 2y^2 + 11x - 5y + \lambda = 0$$

$$\text{xiv. } 6x^2 - 7xy + 16x - 3y^2 - 2y + k = 0$$

Soln

$$\text{viii. } \lambda x^2 + 4xy + y^2 - 4x - 2y - 3 = 0 \quad \text{--- (1)}$$

comparing with $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

$$a = \lambda, h = 2, b = 1, g = -2, f = -1, c = -3$$

~~For pair of straight line~~

for pair of straight line eqn. $\Delta = 0$

$$\text{we know, } \Delta = abc + 2hgf - af^2 - bg^2 - ch^2$$

$$\Rightarrow (\lambda \times 1 \times -3) + (2 \times 2 \times -2 \times -1) - \lambda(-1)^2 - 1(-2)^2 - [(-3) \times 2^2] = 0$$

$$\Rightarrow -\frac{3}{2}\lambda + 8 - \lambda - 4 + 12 = 0$$

$$\Rightarrow -4\lambda = -16 \Rightarrow \lambda = 4 \quad \underline{\underline{\text{Ans}}}$$

$$1x \cdot 2x^2 + xy - y^2 - 2x - 5y + k = 0 \quad (1)$$

$$\therefore a=2, h=\frac{1}{2}, b=-1, g=-1, f=-\frac{5}{2}, c=k$$

$$\therefore \Delta = abc + 2hgf - af^2 - bg^2 - ch^2$$

$$\Rightarrow (2 \times (-1) \times k) + (2 \times \frac{1}{2} \times (-1) \times (-\frac{5}{2})) - [2 \times (-\frac{5}{2})^2] - [(-1) \times (-1)]$$

$$= 0 \Rightarrow \left[\left(\frac{1}{2} \right) \times (20) - \left[k \times \left(\frac{1}{2} \right) \right] \right] = 0$$

$$\Rightarrow -2k + \frac{5}{2} - \frac{25}{2} + 1 - \frac{k}{4} = 0$$

$$\Rightarrow -\frac{9k}{4} - \frac{18}{2} = 0$$

$$\Rightarrow -\frac{9k}{4} = 9$$

$$\Rightarrow -9k = 36$$

$$\Rightarrow k = -4 \quad \underline{\underline{\text{Ans}}}$$

$$\textcircled{10} 6x^2 + xy + ky^2 - 11x + 43y - 35 = 0$$

$$\therefore a = 6, h = \frac{1}{2}, b = k, g = -\frac{11}{2}, f = \frac{43}{2}, c = -35$$

$$\Delta = abc + 2hgf - af^2 - bg^2 - ch^2$$

$$\Rightarrow (6 \times k \times -35) + \left(2 \times \frac{1}{2} \times -\frac{11}{2} \times \frac{43}{2}\right) - \left[6 \times \left(\frac{43}{2}\right)^2\right] - \left[k \times \left(-\frac{11}{2}\right)^2\right] - \left[-35 \times \left(\frac{1}{2}\right)^2\right] = 0$$

$$\Rightarrow -210k - \frac{473}{4} - 2773.5 - \frac{121}{4}k + \frac{35}{4} = 0$$

$$\Rightarrow -\frac{210k}{4} - \frac{12951}{4} = 0$$

$$\Rightarrow -210k = 12951$$

$$\Rightarrow k = \frac{12951}{-210}$$

$$\Rightarrow \frac{961k}{4} - 2883 = 0$$

$$\Rightarrow -\frac{961}{4}k - 2883 = 0$$

$$\Rightarrow -\frac{961k}{4} = 2883$$

$$\Rightarrow -961k = 11532$$

$$\Rightarrow k = \frac{11532}{-961}$$

$$= -12 \quad \underline{\underline{\text{Ans}}}$$

XI. $x^2 - \lambda xy + 2y^2 + 3x - 5y + 2 = 0$

$a=1, h=-\frac{\lambda}{2}, b=2, g=\frac{3}{2}, f=-\frac{5}{2}, c=2$

~~$\Delta = abc + 2hgf - af^2 - bg^2 - ch^2$~~

$\Delta = abc + 2hgf - af^2 - bg^2 - ch^2$

$\Rightarrow [1 \times 2 \times 2] + [2 \times (-\frac{\lambda}{2}) \times \frac{3}{2} \times (-\frac{5}{2})] - 1(-\frac{5}{2})^2 - 2(\frac{3}{2})^2 - 2(-\frac{\lambda}{2})^2 = 0$

$\Rightarrow 4 + \frac{15}{2}\lambda - \frac{25}{4} - \frac{9}{2} - \frac{\lambda^2}{2} = 0$

$\Rightarrow -\frac{\lambda^2}{2} + \frac{15}{2}\lambda - \frac{27}{4} = 0$

$\Rightarrow -2\lambda^2 + 15\lambda - 27 = 0$

$\Rightarrow 2\lambda^2 - 15\lambda + 27 = 0$

$\Rightarrow 2\lambda^2 - 9\lambda - 6\lambda + 27 = 0$

$\Rightarrow 2\lambda^2 - 6\lambda - 6\lambda + 27 = 0$

$\Rightarrow 2\lambda(\lambda - 3) - 9(\lambda - 3) = 0$

$\Rightarrow (\lambda - 3)(2\lambda - 9) = 0$

$\lambda - 3 = 0$

$\lambda = 3 \text{ Ans}$

or $2\lambda - 9 = 0$

or $\lambda = \frac{9}{2} \text{ Ans}$

XII $kxy - 8x + 9y - 12 = 0$

$a=0, h=\frac{k}{2}, b=0, g=-4, f=\frac{9}{2}, c=-12$

$\Delta = abc + 2hfg - af^2 - bg^2 - ch^2$

$\Rightarrow 0 \times 0 \times -12 + 2 \times \frac{k}{2} \times \frac{9}{2} \times (-4) - (-12 \times (\frac{k}{2})^2) = 0$

$\Rightarrow -18k + 3k^2 = 0$

$\Rightarrow 3k^2 - 18k = 0$

$\Rightarrow 3k(k-6) = 0$

$3k = 0, \text{ or } k - 6 = 0$

If $k=0$,
then $xy=0$ it is not possible so $k=6$ Ans

XIII $6x^2 + 2kxy + 12y^2 + 22x + 31y + 20 = 0$

$a=6, h=k, b=12, g=11, f=\frac{31}{2}, c=20$

$\Delta = (6 \times 12 \times 20) + 2k \times 11 \times \frac{31}{2} - 6 \times (\frac{31}{2})^2 - 12 \times (11)^2 - 20k^2 = 0$

$\Rightarrow 1440 + 341k - \frac{2883}{2} - 1452 - 20k^2 = 0$

$\Rightarrow -20k^2 + 341k - \frac{2007}{2} = 0$

$\Rightarrow 40k^2 - 682k + 2007 = 0$

$\therefore k = \frac{171}{20}, \frac{17}{2}$ [using calculator]

XIV $6x^2 - 7xy + 16x - 3y^2 - 2y + k = 0$

$a=6, h=-\frac{7}{2}, b=-3, g=8, f=-1, c=k$

$$\Delta = [6 \times (-3) \times k] + [2 \times -\frac{7}{2} \times 8 \times -1] - [6 \times (-1)^2] - [(-3) \times 8^2] - [k \times (-\frac{7}{2})^2]$$

$$\Rightarrow -18k + 56 - 6 + 192 - \frac{49k}{4} = 0$$

$$\Rightarrow -\frac{121}{4}k + 242 = 0$$

$$\Rightarrow \frac{121k}{4} = 242$$

$$\Rightarrow 121k = 968$$

$$\Rightarrow k = 8 \text{ Ans}$$

XV $12x^2 - 10xy + 2y^2 + 11x - 5y + \lambda = 0$

$a=12, h=-5, b=2, g=\frac{11}{2}, f=-\frac{5}{2}, c=\lambda$

$$[12 \times 2 \times \lambda] + [2 \times (-5) \times \frac{11}{2} \times (-\frac{5}{2})] - [12 \times (-\frac{5}{2})^2] - [2 \times (\frac{11}{2})^2] - \lambda \times (-5)^2 = 0$$

$$\Rightarrow 24\lambda + \frac{275}{2} - 75 - \frac{121}{2} - 25\lambda = 0$$

$$\Rightarrow \cancel{20\lambda} - \lambda + 2 = 0$$

$$\Rightarrow \lambda = 2 \text{ Ans}$$

(21) Show that the two straight lines represent by $ax^2 + 2hxy - ay^2 = 0$ will be perpendicular each other.

Soln: $ax^2 + 2hxy - ay^2 = 0$

comparing $ax^2 + 2hxy + by^2 = 0$

$$a = a, h = h, b = -a$$

We know,

$$\tan \theta = \frac{2\sqrt{h^2 - ab}}{a+b}$$

$$= \frac{2\sqrt{h^2 - a(-a)}}{a+(-a)}$$

$$= \frac{2\sqrt{h^2 + a^2}}{0}$$

$$\tan \theta = \infty$$

$$\Rightarrow \theta = \tan^{-1}(\infty)$$

$$= 90^\circ$$

So $ax^2 + 2hxy - ay^2 = 0$ will be perpendicular each other ~~(obvious)~~

3.9 find the eqn of the bisectors of the angles between the lines represented by

$$a. 12x^2 + 7xy - 10y^2 + 13x + 45y - 35 = 0$$

$$b. 2x^2 + 7xy + 6y^2 + 13x + 22y + 20 = 0$$

Solⁿ a

$$12x^2 + 7xy - 10y^2 + 13x + 45y - 35 = 0 \quad \text{--- (1)}$$

comparing, $a = 12, h = \frac{7}{2}, b = -10, g = \frac{13}{2}, f = \frac{45}{2}$
 $c = -35$

$$\alpha = \frac{hf - bg}{ab - h^2}$$

$$= \frac{\left(\frac{7}{2} \times \frac{45}{2}\right) - \left(-10 \times \frac{13}{2}\right)}{(12 \times (-10)) - \left(\frac{7}{2}\right)^2}$$

$$= \frac{\frac{575}{4} - \frac{529}{4}}{-\frac{529}{4}}$$

$$= -\frac{575}{529}$$

$$= -\frac{25}{23}$$

$$\beta = \frac{hg - af}{ab - h^2}$$

$$= \frac{\left(\frac{7}{2} \times \frac{13}{2}\right) - \left(12 \times \frac{45}{2}\right)}{(12 \times (-10)) - \left(\frac{7}{2}\right)^2}$$

$$= \frac{-\frac{389}{4}}{-\frac{529}{4}}$$

$$= \frac{43}{23}$$

We know, for bisector

$$\frac{(x-\alpha)^2 - (y-\beta)^2}{a-b} = \frac{(x-\alpha)(y-\beta)}{h}$$

$$\Rightarrow \frac{(x - (-\frac{25}{23}))^2 - (y - \frac{43}{23})^2}{12 - (-10)} = \frac{(x + \frac{25}{23})(y - \frac{43}{23})}{\frac{7}{2}}$$

$$\Rightarrow \frac{(\frac{23x+25}{23})^2 - (\frac{23y-43}{23})^2}{22} = \frac{2(\frac{23x+25}{23})(\frac{23y-43}{23})}{7}$$

$$\Rightarrow \frac{7}{23^2} [(23x+25)^2 - (23y-43)^2] = \frac{44}{23^2} [(23x+25)(23y-43)]$$

$$\Rightarrow 7(23x+25)^2 - 7(23y-43)^2 = 44(23x+25)(23y-43)$$

Ans